

General Education Reflection

@svcdown · CPR E 4940 Portfolio Assessment · Oct 6, 2026

When I started at Iowa State, I saw general education courses as requirements to get through before the "real" engineering work. Looking back now as a senior in Cyber Security Engineering, I see it differently. Security is not only a technical problem. Most breaches start with a person who was tricked, a budget that was cut, or a decision made without understanding the risk. My general education courses in communication, psychology, economics, global studies, and leadership gave me the tools to understand those human and societal sides of the problems I will be solving. In this reflection, I explain what I learned in several of those courses and how that knowledge connects to real engineering problems facing society today.

The courses that have helped me most day to day are my communication courses: ENGL 150 and ENGL 250 (Composition I and II), SP CM 212 (Fundamentals of Oral Communication), and ENGL 3140 (Technical Communication). In the composition courses, I learned how to build an argument around evidence and organize my writing for a specific reader. SP CM 212 taught me how to plan a talk, adapt to an audience, and stay clear under pressure. ENGL 3140 pulled these skills together for technical writing, such as instructions, proposals, and reports. In my infrastructure work, these skills matter constantly. A firewall change or a network segmentation plan is useless if I cannot explain why it is needed to a manager who does not know what a VLAN is. Engineers who can translate technical risk into plain language are the ones who get security projects approved and funded.

PSYCH 101 (Introduction to Psychology) is the course I connect most directly to a current problem in my field: social engineering. In that course, I learned how people make quick decisions using mental shortcuts, how stress and authority shape behavior, and how easily memory and perception can be influenced. Attackers exploit exactly these tendencies. Phishing emails create false urgency, impersonate a boss or IT department, and push people to act before they think. Today, generative AI makes this worse, because attackers can now write flawless phishing messages and even clone a voice for a fake phone call. Understanding psychology helps me design better defenses. Instead of blaming users, I can build systems that assume people will make mistakes, such as requiring phishing-resistant multi-factor authentication and designing security training around how people actually behave.

ECON 101 (Principles of Microeconomics) and HD FS 283 (Personal and Family Finance) changed how I think about security decisions. ECON 101 taught me about incentives, opportunity cost, and cost-benefit analysis. Ransomware is a clear example of these ideas in action. It has become a business with suppliers, affiliates, and profit margins, and it keeps growing because paying ransoms makes attacks profitable. Hospitals, schools, and local governments are frequent targets because downtime is so costly that they feel pressure to pay. On the defense side, every organization has a limited budget, so engineers must show that a control is worth its cost. HD FS 283 added a personal side to this by teaching me about budgeting, risk, and protecting financial information. Scams and identity theft cause real financial harm to families, which reminds me that security failures hurt ordinary people, not just companies.

RUS 375 (Russia Today) gave me a global perspective that I did not expect to use in engineering. The course covered Russia's modern politics, media, and relationship with the West. That background helps me understand state-sponsored cyber operations as part of larger geopolitical goals rather than random attacks. Attacks on Ukraine's power grid, the NotPetya malware that spread worldwide, and ongoing disinformation campaigns all show how cyber attacks now serve national strategy. This is a global problem that directly affects critical infrastructure such as power, water, and communications. As an engineer, understanding who the threat actors are and what motivates them helps me prioritize defenses. It also reminds me that the networks I help secure are part of a much larger system that society depends on.

Finally, LDST 3010 (Leadership Theory), which I am taking this semester, has helped me reflect on my experience as a team lead at the ISU IT Solution Center. The course compares leadership approaches such as transformational and situational leadership. I have started to see that the best approach depends on the person and the situation. A new technician needs clear direction, while an experienced one needs trust and room to work. This matters in security because responding to an incident or rolling out a major change is a team effort under pressure. The cybersecurity field also faces a large workforce shortage, so engineers who can mentor and develop others are especially valuable.

My general education courses have shaped me into a more complete engineer. Communication courses taught me to explain risk clearly. Psychology helped me understand social engineering. Economics and finance showed me why ransomware thrives and how to justify security spending. Russia Today gave me a global view of cyber conflict, and Leadership Theory is preparing me to lead teams. Technical skills let me

configure a firewall, but these courses help me understand the people, incentives, and global forces behind the threats. As I graduate and continue my career in infrastructure and security, I will keep drawing on both.